

Enterprise architecture:

Enterprise architecture helps in understanding the structure of enterprise application and can be broken down into 3 logical layers:

- **Presentation layer:** collects data from the user and displays data elements to the external world. it is generally considered as the user interface which includes the part of the software that creates and controls the user interface and validates the users' actions.
- **Business layer:** it's also known as logical layer which helps an application to work, and handle processing of business logic.
- **Data storage and access:**
 - this layer helps business application to read and store data.

Enterprise architecture type:

1. one tier architecture
2. two tier architecture
3. 3 tier architecture
4. N tier architecture

Java EE (enterprise edition) architecture:

Java ee is based on the multi-tier distributed application model in which the applications logic is segregated into various application components. a java ee application can be 3 tier or 4 tier, but generally they are considered to be three tier as they are located over 3 different locations that are client machine , java ee server and database server. this implies that the application logic is divided at different machines such as client , java ee server and the database at the backend.

Exploring the HTTP Protocol:

http is a stateless (means each request is considered as the new request , in other words server doesn't recognize the user by default), connection less (in which http client that is a browser initiates the http request and after the request is send the client disconnects from server and wait for

the response , application level , communication protocol used to transfer information over a network . it also defines the type of requests that client can send to servers ,and the type of responses that server can send to client . each request has a url which identifies a web component or a static object such as an html page or an image file. when http is used for communication between client and server in a web application , the web server converts the http request, send by the client , to an http request object . next the httpRequest object is delivered to web component that is identified by the request url. Further the web component sends the HTTPResponseobject which is then converted into the http response by the application server and finally send to the client.

HTTP requests:

(GET | POST | HEAD | TRACE | PUT | DELETE | OPTIONS)

ANATOMY OF GET REQUEST:

THE QUERY string (name value pair) is sent inside the url of a get request

For example:

<http://www.conclave.com/login.jsp?user=sumangal&pwd=1234>

scheme: <http://www>

second level domain: conclave

top level domain: com

path to the source on web server: login.jsp

parameters to the server (query string) : user=sumangal&pwd=1234

ANATOMY OF POST REQUEST:

<http://www.conclave.com/login.jsp>

scheme: <http://www>

second level domain: conclave

top level domain: com

path to the source on web server: login.jsp

SERVLET:

A servlet is a small java program that runs within a web server . servlet receive and response to requests from web clients , usually across http . it provides high level component based platform independent and server independent standard to develop web applications in java .

A servlet is secured and portable as it runs on jvm embedded with web server and cannot operate outside the domain of web server . servlets are java components that are used to create dynamic web pplications . they can run on any java enabled platform and are usually designed to process http request such as get and post .

SERVLET API:

INTERFACES IN JAVAX. SERVLETPACKAGE :

1. **servelet**
2. **ServeletRequest**
3. **ServeletResponse**
4. **RequestDispatcher**
5. **ServeletConfig**
6. **ServeletContext**
7. **SingleThreadModel**
8. **Filter**
9. **filterConfig**
10. **filterChain**
11. **servletRequestListener**
12. **servletRequestAttributelistener**
13. **servletContextListener**
14. **servletContextAttributeListener**

CLASSES IN JAVAX. SERVLETPACKAGE

1. **Genericservlet**
2. **servletInputStream**
3. **servletOutputStream**

4. servletRequestWrapper
5. servletResponseWrapper
6. servletRequestEvent
7. servletContextEvent
8. servletRequestAttributeEvent
9. servletContextAttributeEvent
10. servletException
11. unavailableException

INTERFACES IN JAVAX. SERVLET.HTTPPACKAGE:

1. **httpServletRequest**
2. **httpServletResponse**
3. **httpSession**
4. **httpSessionListener**

CLASSES IN JAVAX. SERVLET.HTTPPACKAGE:

1. httpServlet
2. Cookie
3. httpServletRequestWrapper
4. httpServletResponseWrapper
5. httpSessionEvent